MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED LOW-POWER SCHOTTKY, TTL, OCTAL BUFFER GATES WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, advanced low-power Schottky, octal bus buffer gates with three-state outputs. Two product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Inverting octal buffer gate (inverting control inputs)
02	Noninverting octal buffer gate (complementary control inputs)
03	Noninverting octal buffer gates (inverting control inputs)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
X	C-2A (20-terminal, .350" x .350"), square chip carrier package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to 7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to 7.0 V dc
Storage temperature range	-65°C to 150°C
Maximum power dissipation, (P_D) per device 1/	
Device type 01	137.5 mW
Device types 02, 03	165 mW
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case θ_{JC}:	
Cases R, S, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature T_J 2/	175°C

1/ Must withstand the added P_D due to short circuit test (e.g. I_O).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBR-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

1.4 Recommended operating conditions:

Supply voltage - - - - - - - - - - - - - - - -	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH}) - - -	2.0 V
Maximum low level input voltage (V_{IL})- - - -	0.8 V except at (T_C = +125°C, 0.7 V)
Case operating temperature range (T_C)- - - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$ unless otherwise specified		Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH1}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, I_{OH} = -3 mA V_{IH} = 2.0 V (02, 03 only) T_C = 125°C, V_{IC} = 0.7 V		All	2.4		V
High level output voltage	V_{OH2}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, I_{OH} = -12 mA V_{IH} = 2.0 V (02, 03 only) T_C = 125°C, V_{IC} = 0.7 V		All	2.0		V
Low level output voltage	V_{OL1}	V_{CC} = 4.5 V, I_{OL} = 12 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V T_C = 125°C, V_{IC} = 0.7 V		All		0.4	V
Low level output voltage	V_{OL2}	V_{CC} = 4.5 V, I_{OL} = 18 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V T_C = 125°C, V_{IC} = 0.7 V		All		0.45	V
Off-state (high-impedance state) output current	I_{OZH}	V_{CC} = 5.5 V, V_{OUT} = 2.7 V, V_{IH} = 2.0 V		All		20	µA
Off-state (low-impedance state) output current	I_{OZL}	V_{CC} = 5.5 V, V_{OUT} = 0.4 V, V_{IH} = 2.0 V		All		-20	µA
Low level input current (all inputs)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V		All	0	-100	µA
High level input current (all inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		All		20	µA
High level input current (all inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V		All		100	µA
Output current 1/	I_O	V_{CC} = 5.5 V, V_{OUT} = 2.25 V		All	-30	-112	mA
Supply current	I_{CCH}	V_{CC} = 5.5 V	Outputs high	01 02 03		10 17 15	mA
	I_{CCL}		Outputs low	01 02 03		23 28 24	
	I_{CCZ}		All outputs disabled	01 02 03		25 32 27	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	All		-1.5	V
Propagation delay time (low-to-high level)	t_{PLH1}	C_L = 50 pF, R_L = 500 ohms, V_{CC} = 5.0 V	01	3	12	ns
			02	3	14	
			03	3	13	
Propagation delay time (high-to-low level)	t_{PHL1}		01	2	11	ns
			02,03	3	13	
Output enable time to low level	t_{PZL}		01	5	20	ns
			02,03	7	25	
Output enable time to high level	t_{PZH}		01	5	15	ns
			02,03	7	25	
Output disable time from low level	t_{PLZ}		02	3	20	ns
			01,03	3	18	
Output disable time from high level	t_{PHZ}		01,02 03	2	12	ns

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8, 10,11	1,2,3,7,8, 9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,9, 10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

* PDA applies to subgroup 1 (see 4.2c.).

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D using the circuit shown on figure 3, or equivalent.

2. T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

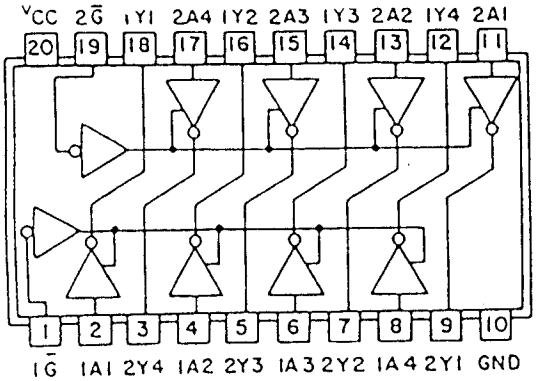
4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

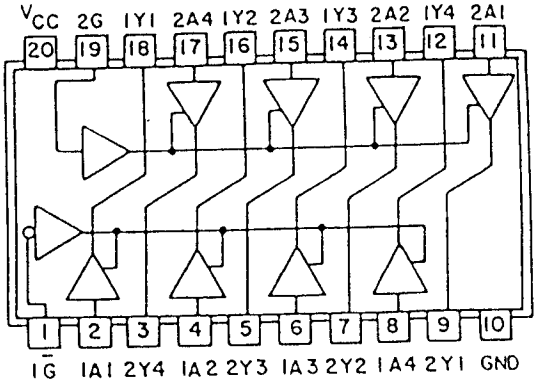
Device type 01

Cases R and S



Device type 02

Cases R and S



Device type 03

Cases R and S

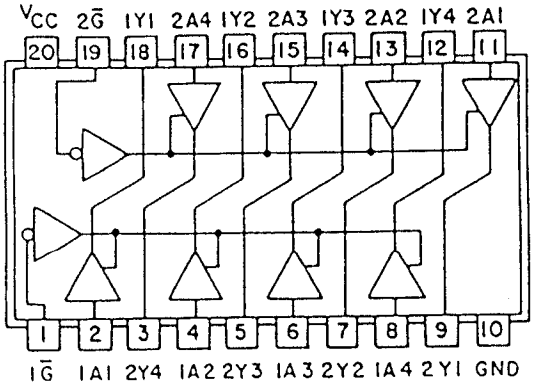
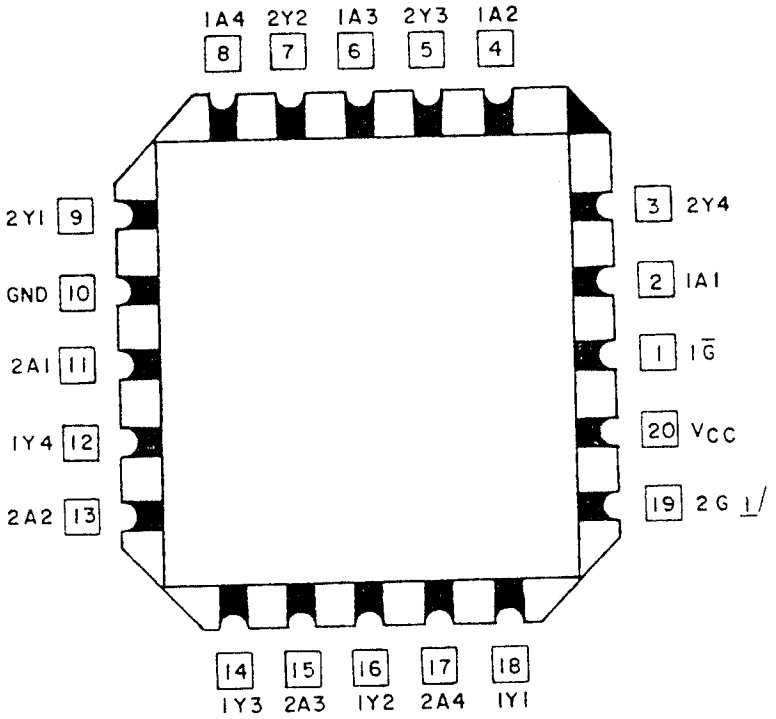


FIGURE 1. Logic diagrams and terminal connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

Device types 01, 02, and 03

Cases 2 and X



TOP VIEW

1/ For 01 and 03 pin 19 = $2\bar{G}$

FIGURE 1. Logic diagrams and terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

Device type 01

Cases 2 and X

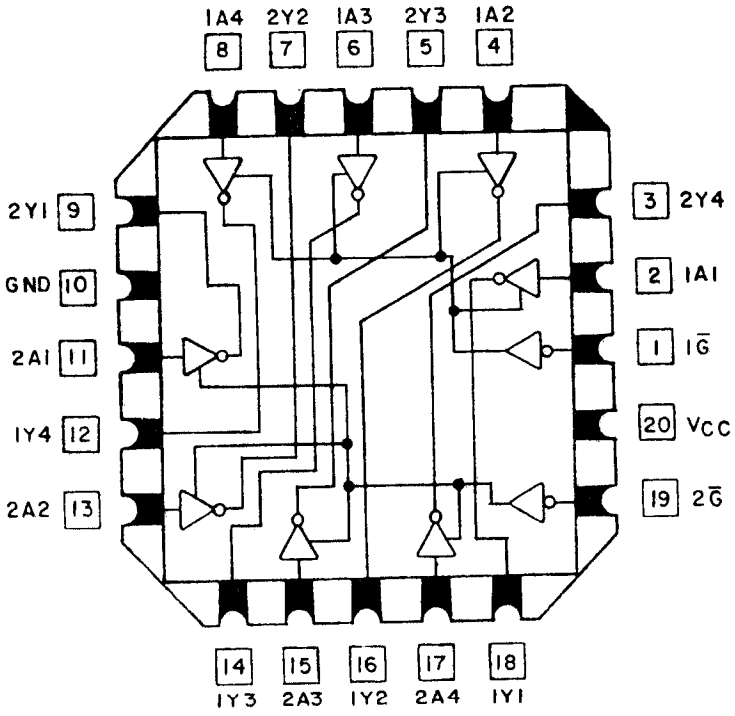
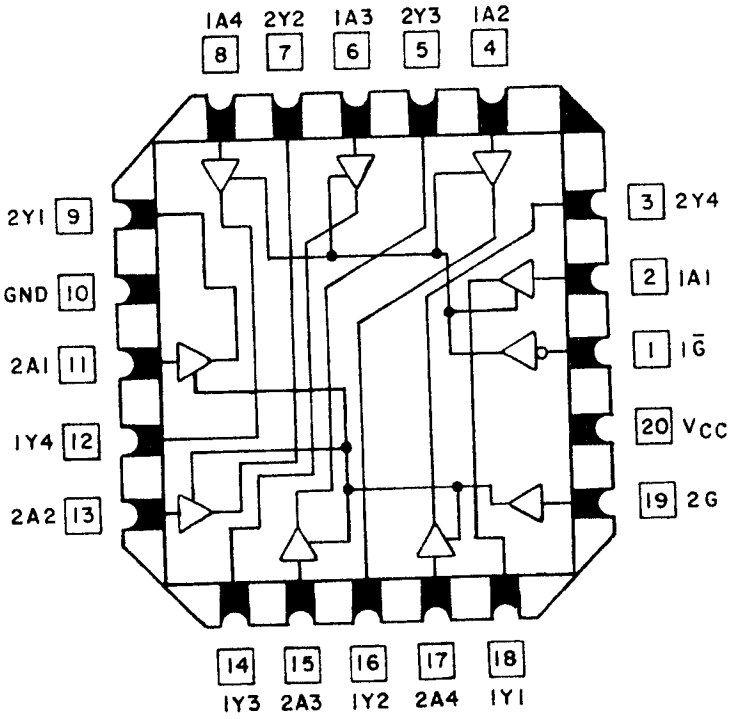


FIGURE 1. Logic diagrams and terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

Device type 02

Cases 2 and X



Device type 03

Cases 2 and X

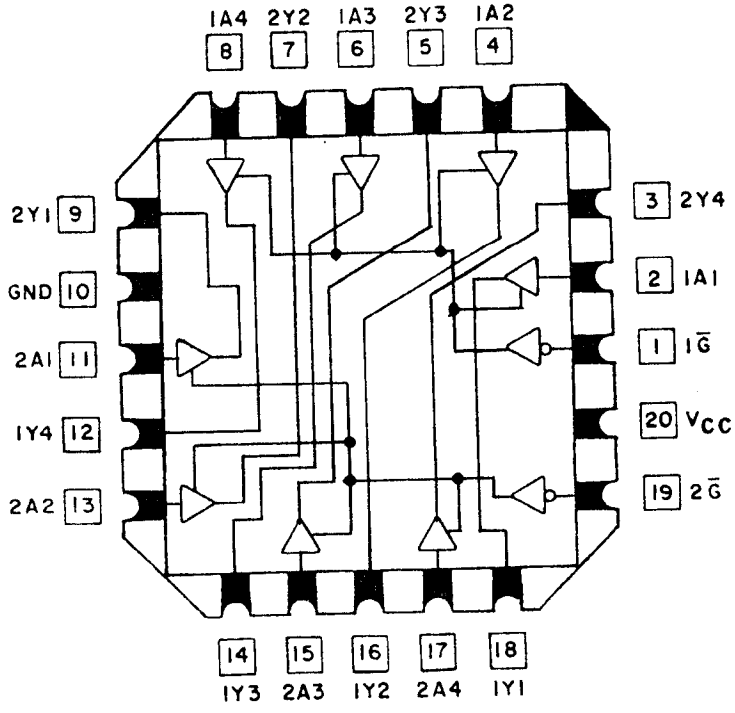


FIGURE 1. Logic diagrams and terminal connections - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

Device type 01

Inputs		Out
$\overline{G}$	A	Y
L	H	L
L	L	H
H	X	Z

Device type 02

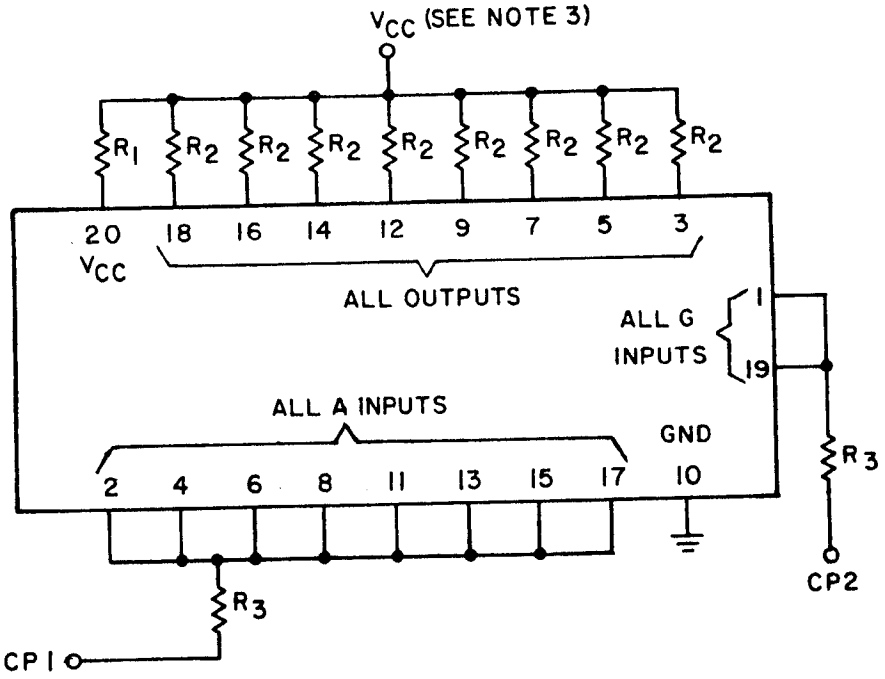
Inputs				Outputs	
$1\overline{G}$	2G	1A	2A	1Y	2Y
L	H	H	H	H	H
L	H	L	L	L	L
H	L	X	X	Z	Z

Device type 03

Inputs		Out
$\overline{G}$	A	Y
L	L	L
L	H	H
H	X	Z

FIGURE 2. Truth tables.

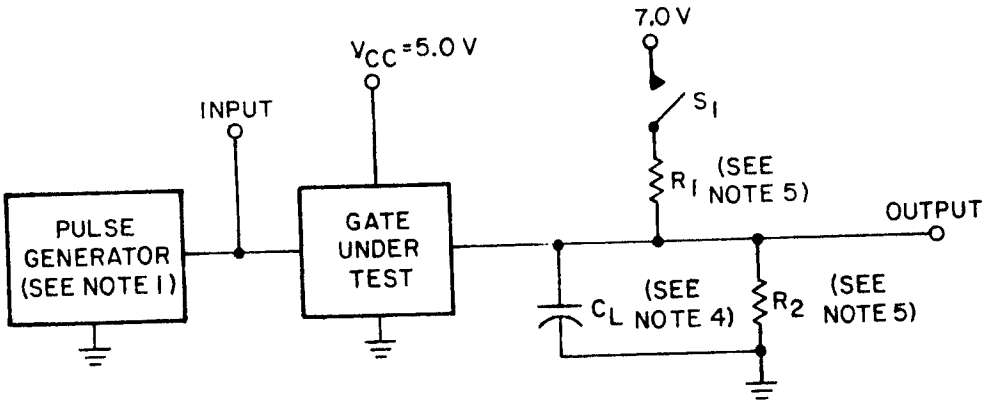
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.



NOTES:
1. CP1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%, V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. CP2 = 50 kHz ±50% square wave, otherwise as described in note 1 above.
3. R_1 = 10 Ω maximum; R_2 = 470 Ω ±5%; R_3 = 27 Ω maximum.
4. V_{CC} shall be high enough to insure 5.0 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life-test circuit for device types 01, 02 and 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

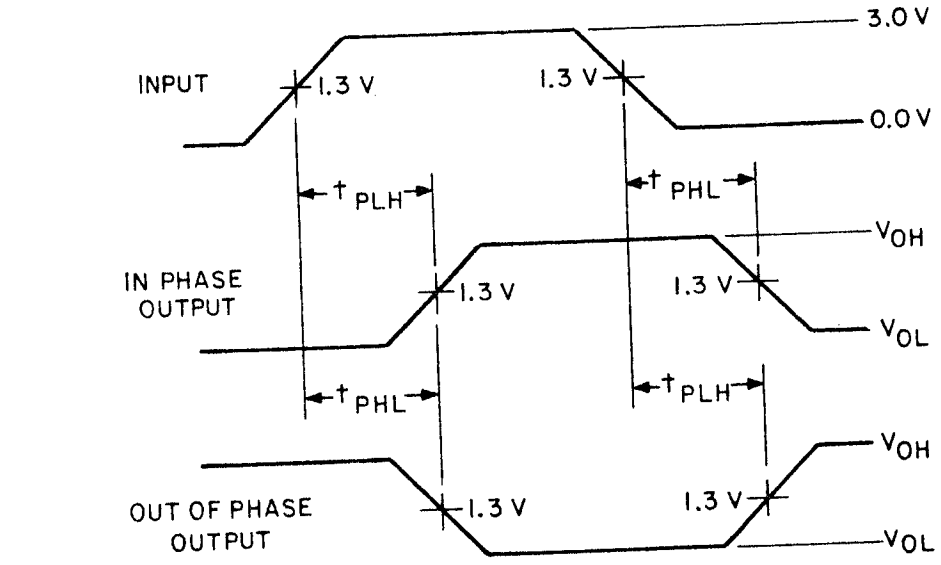


TEST CIRCUIT

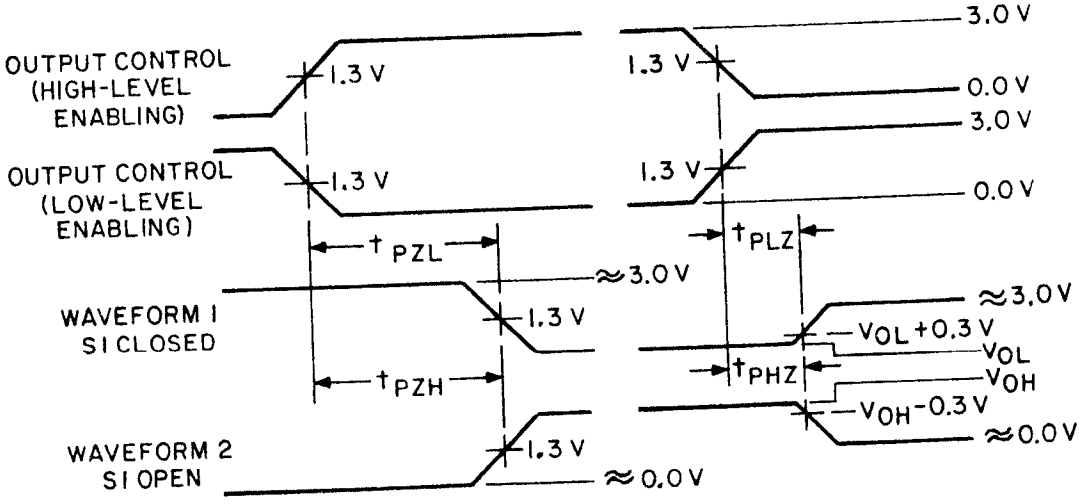
NOTES:
1. All input pulses are supplied by generators having the following characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns, PRR ≤ 1 MHz, $Z_{OUT} \approx 50\ \Omega$.
2. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
   Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
3. In the examples above, the phase relationships between inputs and outputs have been chosen arbitrarily.
4. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
5. $R_1 = R_2 = 499\ \Omega \pm 1\%$.
6. Voltage measurements are made with respect to network ground terminal.

FIGURE 4. Switching time test circuit for all device types.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES, THREE-STATE OUTPUTS

FIGURE 4. Switching time test circuit for all device types - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2 Test no.	1 $\overline{1G}$	2 1A1	3 2Y4	4 1A2	5 2Y3	6 1A3	7 2Y2	8 1A4	9 2Y1	10 GND	11 2A1	12 1Y4	13 2A2	14 1Y3	15 2A3	16 1Y2	17 2A4	18 1Y1	19 $\overline{2G}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	$V_{OH}1$	3006	1	0.8 V	0.8 V		0.8 V		0.8 V		0.8 V		GND				-3 mA		-3 mA		-3 mA		4.5 V	1Y1	2.4		V
	"	"	2	"									"										"	1Y2	"		"
	"	"	3	"									"										"	1Y3	"		"
	"	"	4	"									"										"	1Y4	"		"
	"	"	5	"									"										"	2Y1	"		"
	"	"	6	"									"										"	2Y2	"		"
	"	"	7	"									"										"	2Y3	"		"
	"	"	8	"		-3 mA		-3 mA		-3 mA		-3 mA	"	0.8 V	-3 mA	0.8 V		0.8 V		0.8 V		0.8 V	"	2Y4	"		"
	$V_{OH}2$	"	9	0.8 V	0.8 V		0.8 V		0.8 V		0.8 V		"				-12 mA		-12 mA		-12 mA		"	1Y1	2.0		"
	"	"	10	"									"										"	1Y2	"		"
	"	"	11	"									"										"	1Y3	"		"
	"	"	12										"										"	1Y4	"		"
	"	"	13										"										"	2Y1	"		"
	"	"	14										"										"	2Y2	"		"
	"	"	15										"										"	2Y3	"		"
	"	"	16			-12 mA		-12 mA		-12 mA		-12 mA	"	0.8 V	-12 mA	0.8 V		0.8 V		0.8 V		0.8 V	"	2Y4	"		"
	$V_{OL}1$	3007	17	0.8 V	2.0 V		2.0 V		2.0 V		2.0 V		"				12 mA		12 mA		12 mA		"	1Y1		0.4	"
	"	"	18	"									"										"	1Y2		"	"
	"	"	19	"									"										"	1Y3		"	"
	"	"	20										"										"	1Y4		"	"
	"	"	21										"										"	2Y1		"	"
	"	"	22										"										"	2Y2		"	"
	"	"	23										"										"	2Y3		"	"
	"	"	24			12 mA		12 mA		12 mA		12 mA	"	2.0 V	12 mA	2.0 V		2.0 V		2.0 V		0.8 V	"	2Y4		"	"
	$V_{OL}2$	"	25	0.8 V	2.0 V		2.0 V		2.0 V		2.0 V		"				18 mA		18 mA		18 mA		"	1Y1		0.45	"
	"	"	26	"									"										"	1Y2		"	"
	"	"	27	"									"										"	1Y3		"	"
	"	"	28										"										"	1Y4		"	"
	"	"	29										"										"	2Y1		"	"
	"	"	30										"										"	2Y2		"	"
	"	"	31										"										"	2Y3		"	"
	"	"	32			18 mA		18 mA		18 mA		18 mA	"	2.0 V	18 mA	2.0 V		2.0 V		2.0 V		0.8 V	"	2Y4		"	"
	I_{OZH}		33	2.0 V									"								2.7 V		5.5 V	1Y1		20	µA
	"		34	"									"										"	1Y2		"	"
	"		35	"									"										"	1Y3		"	"
	"		36										"										"	1Y4		"	"
	"		37										"										"	2Y1		"	"
	"		38										"										"	2Y2		"	"
	"		39										"										"	2Y3		"	"
	"		40			2.7 V		2.7 V		2.7 V		2.7 V	"	2.0 V	2.7 V	2.0 V		2.7 V		2.7 V		2.0 V	"	2Y4		"	"
	I_{OZL}		41	2.0 V									"								0.4 V		5.5 V	1Y1		-20	µA
	"		42	"									"										"	1Y2		"	"
	"		43	"									"										"	1Y3		"	"
	"		44										"										"	1Y4		"	"
	"		45										"										"	2Y1		"	"
	"		46										"										"	2Y2		"	"
	"		47										"										"	2Y3		"	"
	"		48			0.4 V		0.4 V		0.4 V		0.4 V	"	0.4 V	0.4 V	0.4 V		0.4 V		0.4 V		2.0 V	"	2Y4		"	"
	I_{IL}	3009	49	0.4 V	0.4 V		0.4 V		0.4 V		0.4 V		"										"	$\overline{1G}$	1/	1/	"
	"	"	50										"										"	1A1	"	"	"
	"	"	51										"										"	1A2	"	"	"
	"	"	52										"										"	1A3	"	"	"
	"	"	53										"										"	1A4	"	"	"
	"	"	54										"										"	2A1	"	"	"
	"	"	55										"										"	2A2	"	"	"
	"	"	56										"										"	2A3	"	"	"
	"	"	57										"										"	2A4	"	"	"
	"	"	58										"										"	$\overline{2G}$	"	"	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2 / Test no.	1 / 1$\overline{G}$	2 / 1A1	3 / 2Y4	4 / 1A2	5 / 2Y3	6 / 1A3	7 / 2Y2	8 / 1A4	9 / 2Y1	10 / GND	11 / 2A1	12 / 1Y4	13 / 2A2	14 / 1Y3	15 / 2A3	16 / 1Y2	17 / 2A4	18 / 1Y1	19 / 2$\overline{G}$	20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	I_{IH1}	3010	59	2.7 V									GND									5.5 V	1$\overline{G}$		20	µA
	"	"	60		2.7 V								"									"	1A1		"	"
	"	"	61				2.7 V						"									"	1A2		"	"
	"	"	62						2.7 V				"									"	1A3		"	"
	"	"	63								2.7 V		"									"	1A4		"	"
	"	"	64										"	2.7 V								"	2A1		"	"
	"	"	65										"			2.7 V						"	2A2		"	"
	"	"	66										"					2.7 V				"	2A3		"	"
	"	"	67										"							2.7 V		"	2A4		"	"
	"	"	68										"								2.7 V	"	2$\overline{G}$		"	"
	I_{IH2}	"	69	7.0 V									"									"	1$\overline{G}$		100	"
	"	"	70		7.0 V								"									"	1A1		"	"
	"	"	71				7.0 V						"									"	1A2		"	"
	"	"	72						7.0 V				"									"	1A3		"	"
	"	"	73								7.0 V		"									"	1A4		"	"
	"	"	74										"	7.0 V								"	2A1		"	"
	"	"	75										"			7.0 V						"	2A2		"	"
	"	"	76										"					7.0 V				"	2A3		"	"
	"	"	77										"							7.0 V		"	2A4		"	"
	"	"	78										"								7.0 V	"	2$\overline{G}$		"	"
	$I_{O_{\overline{H}}}$ 2/	3011	79	GND	GND		GND						"									"	1Y1	3/ -30	3/ -112	mA
	"	"	80	"					GND				"									"	1Y2	"	"	"
	"	"	81	"							GND		"	GND								"	1Y3	"	"	"
	"	"	82	"									"		2.25 V							"	1Y4	"	"	"
	"	"	83										"			GND						"	2Y1	"	"	"
	"	"	84										"					GND				"	2Y2	"	"	"
	"	"	85										"							GND		"	2Y3	"	"	"
	"	"	86										"									"	2Y4	"	"	"
	I_{CCH}	3005	87	GND	GND		GND		GND		GND		"	GND		GND		GND		GND		"	V_{CC}		10	"
	I_{CCL}	"	88	GND	5.5 V		5.5 V		5.5 V		5.5 V		"	5.5 V		5.5 V		5.5 V		5.5 V		"	V_{CC}		23	"
	I_{CCZ}	"	89	5.5 V									"									"	V_{CC}		25	"
	V_{IC}		90	-18 mA									"									4.5 V	1$\overline{G}$		-1.5	V
	"		91		-18 mA								"									"	1A1		"	"
	"		92				-18 mA						"									"	1A2		"	"
	"		93						-18 mA				"									"	1A3		"	"
	"		94								-18 mA		"									"	1A4		"	"
	"		95										"	-18 mA								"	2A1		"	"
	"		96										"			-18 mA						"	2A2		"	"
	"		97										"					-18 mA				"	2A3		"	"
	"		98										"							-18 mA		"	2A4		"	"
	"		99										"								-18 mA	"	2$\overline{G}$		"	"
2 T_C = +125°C	Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C and V_{IC} tests are omitted and V_{IL} = +0.7 V.																										
3 T_C = -55°C	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																										
7 4/ T_C = 25°C	Truth table tests	3014 3014	100 101	B B	B A	H L	B A	H L	B A	H L	B A	H L	GND GND	B A	H L	B A	H L	B A	H L	B A	H L	B A	5.0 V 5.0 V	5/				
8	Same tests and terminal conditions as subgroup 7 except T_C = +125°C and T_C = -55°C.																										

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	1$\overline{G}$	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2$\overline{G}$	V_{CC}		Min	Max	
9 T_C = 25°C	t_{PLH1}	3003 See fig 6	102 103 104 105 106 107 108 109	GND " "	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND " " " " " " "	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND " "	5.0 V " " " " " " "	1A1 1Y1 1A2 1Y2 1A3 1Y3 1A4 1Y4 2A1 2Y1 2A2 2Y2 2A3 2Y3 2A4 2Y4	3 " " " " " " "	9 " " " " " " "	ns " " " " " " "
	t_{PHL1}	"	110 111 112 113 114 115 116 117	GND " "	IN	OUT	IN	OUT	IN	OUT	IN	OUT	" " " " " " " "	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND " "	" " " " " " " "	1A1 1Y1 1A2 1Y2 1A3 1Y3 1A4 1Y4 2A1 2Y1 2A2 2Y2 2A3 2Y3 2A4 2Y4	2 " " " " " " "	9 " " " " " " "	" " " " " " " "
	t_{PZL}	"	118 119 120 121 122 123 124 125	IN " "	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	" " " " " " " "	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	IN " "	" " " " " " " "	1$\overline{G}$ 1Y1 1$\overline{G}$ 1Y2 1$\overline{G}$ 1Y3 1$\overline{G}$ 1Y4 2$\overline{G}$ 2Y1 2$\overline{G}$ 2Y2 2$\overline{G}$ 2Y3 2$\overline{G}$ 2Y4	5 " " " " " " "	18 " " " " " " "	" " " " " " " "
	t_{PZH}	"	126 127 128 129 130 131 132 133	IN " "	GND	OUT	GND	OUT	GND	OUT	GND	OUT	" " " " " " " "	GND	OUT	GND	OUT	GND	OUT	GND	OUT	IN " "	" " " " " " " "	1$\overline{G}$ 1Y1 1$\overline{G}$ 1Y2 1$\overline{G}$ 1Y3 1$\overline{G}$ 1Y4 2$\overline{G}$ 2Y1 2$\overline{G}$ 2Y2 2$\overline{G}$ 2Y3 2$\overline{G}$ 2Y4	5 " " " " " " "	13 " " " " " " "	" " " " " " " "
	t_{PLZ}	"	134 135 136 137 138 139 140 141	IN " "	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	" " " " " " " "	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	4.5 V	OUT	IN " "	" " " " " " " "	1$\overline{G}$ 1Y1 1$\overline{G}$ 1Y2 1$\overline{G}$ 1Y3 1$\overline{G}$ 1Y4 2$\overline{G}$ 2Y1 2$\overline{G}$ 2Y2 2$\overline{G}$ 2Y3 2$\overline{G}$ 2Y4	3 " " " " " " "	12 " " " " " " "	" " " " " " " "
	t_{PHZ}	"	142 143 144 145 146 147 148 149	IN " "	GND	OUT	GND	OUT	GND	OUT	GND	OUT	" " " " " " " "	GND	OUT	GND	OUT	GND	OUT	GND	OUT	IN " "	" " " " " " " "	1$\overline{G}$ 1Y1 1$\overline{G}$ 1Y2 1$\overline{G}$ 1Y3 1$\overline{G}$ 1Y4 2$\overline{G}$ 2Y1 2$\overline{G}$ 2Y2 2$\overline{G}$ 2Y3 2$\overline{G}$ 2Y4	2 " " " " " " "	10 " " " " " " "	" " " " " " " "

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	1$\overline{G}$	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	2$\overline{G}$	V_{CC}		Min	Max	
10 T_C = 125°C	t_{PLH1} t_{PHL1} t_{PZL} t_{PZH} t_{PLZ} t_{PHZ}	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																							3 2 5 5 3 2	12 11 20 15 18 12	ns " " " " "
11 T_C = -55°C	Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C.																										

1/ I_{IL} limits shall be as follows:

Symbol	Min/max limits (μA) for circuit		
	A	B	C
I_{IL}	0/-100	0/-100	0/-100

2/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

3/ I_O limits for circuit C shall be -30 to -110 mA.

4/ A ≥2.0 V and B ≤0.8 V.

5/ Output voltages shall be either:

a. H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or

b. H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device types 02 and 03. 1/
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	1/	V_{CC}		Min	Max	
1 T_C = 25°C	V_{OH1}	3007	1	0.8 V	2.0 V		2.0 V		2.0 V		2.0 V		GND	2.0 V									4.5 V	1Y1	2.4		V
			2	"									"		-3 mA		-3 mA						"	1Y2	"		
			3	"									"			2.0 V							"	1Y3	"		
			4							-3 mA			"					2.0 V					"	1Y4	"		
			5					-3 mA					"						-3 mA					"	2Y1	"		
			6				2.0 V						"							2.0 V			"	2Y2	"		
			7			-3 mA							"								-3 mA	2/	"	2Y3	"		
			8										"										"	2Y4	"		
	V_{OH2}	"	9	0.8 V	2.0 V								"										"	1Y1	2.0		"
			10	"									"										"	1Y2	"		
			11	"									"										"	1Y3	"		
			12										"								-12 mA			"	1Y4	"		
			13										"											"	2Y1	"		
			14										"											"	2Y2	"		
			15										"									2/		"	2Y3	"		
			16			-12 mA							"											"	2Y4	"		
	V_{OL1}	3006	17	0.8 V	0.8 V		0.8 V		0.8 V		0.8 V		"	0.8 V										1Y1		0.4	V
			18	"									"											1Y2		"	
			19	"									"											1Y3		"	
			20										"								12 mA				1Y4		"	
			21										"												2Y1		"	
			22										"												2Y2		"	
			23										"									2/			2Y3		"	
			24			12 mA		12 mA		12 mA		12 mA	"		12 mA		12 mA		12 mA						2Y4		"	
	V_{OL2}	"	25	0.8 V	0.8 V		0.8 V		0.8 V		0.8 V		"	0.8 V		0.8 V		0.8 V		0.8 V				1Y1		0.45	V
			26	"									"											1Y2		"	
			27	"									"											1Y3		"	
			28										"								18 mA				1Y4		"	
			29										"												2Y1		"	
			30										"												2Y2		"	
			31										"									2/			2Y3		"	
			32			18 mA		18 mA		18 mA		18 mA	"		18 mA		18 mA		18 mA						2Y4		"	
	I_{OZH}		33	2.0 V									"								2.7 V		5.5 V	1Y1		20	μA
			34	"									"										"	1Y2		"	
			35	"									"										"	1Y3		"	
			36										"										"	1Y4		"	
			37										"										"	2Y1		"	
			38										"										"	2Y2		"	
			39										"									3/	"	2Y3		"	
			40			2.7 V		2.7 V		2.7 V		2.7 V	"		2.7 V		2.7 V		2.7 V				"	2Y4		"	
	I_{OZL}		41	2.0 V									"								0.4 V		5.5 V	1Y1		-20	μA
			42	"									"										"	1Y2		"	
			43	"									"										"	1Y3		"	
			44										"										"	1Y4		"	
			45										"										"	2Y1		"	
			46										"										"	2Y2		"	
			47										"									3/	"	2Y3		"	
			48			0.4 V		0.4 V		0.4 V		0.4 V	"		0.4 V		0.4 V		0.4 V				"	2Y4		"	
	I_{IL}	3009	49	0.4 V	0.4 V		0.4 V		0.4 V		0.4 V		"	0.4 V		0.4 V		0.4 V		0.4 V		0.4 V	5.5 V	1G	4/	4/	V
			50	"									"										"	1A1	"	"	
			51	"									"										"	1A2	"	"	
			52										"										"	1A3	"	"	
			53										"										"	1A4	"	"	
			54										"										"	2A1	"	"	
			55										"										"	2A2	"	"	
			56										"										"	2A3	"	"	
			57										"										"	2A4	"	"	
			58										"										"	2G	"	"	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for devices types 02 and 03 1/ - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2 / Test no.	1 / 1G	2 / 1A1	3 / 2Y4	4 / 1A2	5 / 2Y3	6 / 1A3	7 / 2Y2	8 / 1A4	9 / 2Y1	10 / GND	11 / 2A1	12 / 1Y4	13 / 2A2	14 / 1Y3	15 / 2A3	16 / 1Y2	17 / 2A4	18 / 1Y1	19 / 1/	20 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = 25°C	I_{IH1}	3010	59 60 61 62 63 64 65 66 67 68	2.7 V	2.7 V		2.7 V		2.7 V		2.7 V		GND " " " " " " " "	2.7 V		2.7 V		2.7 V		2.7 V		2.7 V	5.5 V " " " " " " " "	1G 1A1 1A2 1A3 1A4 2A1 2A2 2A3 2A4 2G		20 " " " " " " " "	µA " " " " " " " "
	I_{IH2}	"	69 70 71 72 73 74 75 76 77 78	7.0 V	7.0 V		7.0 V		7.0 V		7.0 V		" " " " " " " " "	7.0 V		7.0 V		7.0 V		7.0 V		7.0 V	" " " " " " " " "	1G 1A1 1A2 1A3 1A4 2A1 2A2 2A3 2A4 2G		100 " " " " " " " "	" " " " " " " " "
	I_{O} 5/	3011	79 80 81 82 83 84 85 86	GND " "	5.5 V	2.25 V	5.5 V	2.25 V	5.5 V	2.25 V	5.5 V	2.25 V	" " " " " " " "	5.5 V	2.25 V	5.5 V	2.25 V	5.5 V	2.25 V	5.5 V	2.25 V	7/ " "	" " " " " " " "	1Y1 1Y2 1Y3 1Y4 2Y1 2Y2 2Y3 2Y4	6/ -30 " " " " " "	6/ -112 " " " " " "	mA " " " " " " "
	I_{CCH} I_{CCL} I_{CCZ}	3005 " "	87 88 89	GND GND 5.5 V	5.5 V GND		5.5 V GND		5.5 V GND		5.5 V GND		" " "	5.5 V GND		5.5 V GND		5.5 V GND		5.5 V GND		" " 8/	" " "	V_{CC} V_{CC} V_{CC}		9/ " "	" " "
	V_{IC}		90 91 92 93 94 95 96 97 98 99	-18 mA	-18 mA		-18 mA		-18 mA		-18 mA		" " " " " " " " "	-18 mA		-18 mA		-18 mA		-18 mA		-18 mA	4.5 V " " " " " " " "	1G 1A1 1A2 1A3 1A4 2A1 2A2 2A3 2A4 2G		-1.5 " " " " " " " "	V " " " " " " " "
2 T_C = +125°C	Same tests, terminal conditions and limits as subgroup 1 except T_C = +125°C and V_{IC} tests are omitted.																									
3 T_C = -55°C	Same tests, terminal conditions and limits as subgroup 1 except T_C = -55°C and V_{IC} tests are omitted.																									
7 T_C = 25°C 10/	Truth table tests	3014 3014	100 101	B B	B A	L H	B A	L H	B A	L H	B A	L H	GND GND	B A	L H	B A	L H	B A	L H	B A	L H	11/ 11/	5.0 V 5.0 V	12/			
8	Same tests and terminal conditions as subgroup 7 except T_C = +125°C and T_C = -55°C.																									

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device types 02 and 03 1/ - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	1/	V_{CC}		Min	Max	
9 T_C = 25°C	t_{PLH1}	3003 See fig 6	102 103 104 105 106 107 108 109	GND " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	GND " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	7/ " " " " " " "	5.0 V " " " " " " "	1A1 1Y1 1A2 1Y2 1A3 1Y3 1A4 1Y4 2A1 2Y1 2A2 2Y2 2A3 2Y3 2A4 2Y4	3 " " " " " " "	13/ " " " " " " "	ns " " " " " " "
	t_{PHL1}	" " " " " " " "	110 111 112 113 114 115 116 117	GND " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	" " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	IN " " " " " " "	OUT " " " " " " "	7/ 7/ " " " " " "	" " " " " " " "	1A1 1Y1 1A2 1Y2 1A3 1Y3 1A4 1Y4 2A1 2Y1 2A2 2Y2 2A3 2Y3 2A4 2Y4	3 " " " " " " "	10 " " " " " " "	" " " " " " " "
	t_{PZL}	" " " " " " " "	118 119 120 121 122 123 124 125	IN " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	IN " " " " " " "	" " " " " " " "	1G 1Y1 1G 1Y2 1G 1Y3 1G 1Y4 2G 2Y1 2G 2Y2 2G 2Y3 2G 2Y4	7 " " " " " " "	14/ " " " " " " "	" " " " " " " "
	t_{PZH}	" " " " " " " "	126 127 128 129 130 131 132 133	IN " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	IN " " " " " " "	" " " " " " " "	1G 1Y1 1G 1Y2 1G 1Y3 1G 1Y4 2G 2Y1 2G 2Y2 2G 2Y3 2G 2Y4	7 " " " " " " "	15/ " " " " " " "	" " " " " " " "
	t_{PLZ}	" " " " " " " "	134 135 136 137 138 139 140 141	IN " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	" " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	GND " " " " " " "	OUT " " " " " " "	IN " " " " " " "	" " " " " " " "	1G 1Y1 1G 1Y2 1G 1Y3 1G 1Y4 2G 2Y1 2G 2Y2 2G 2Y3 2G 2Y4	3 " " " " " " "	10 " " " " " " "	" " " " " " " "
	t_{PHZ}	" " " " " " " "	142 143 144 145 146 147 148 149	IN " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	" " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	4.5 V " " " " " " "	OUT " " " " " " "	IN " " " " " " "	" " " " " " " "	1G 1Y1 1G 1Y2 1G 1Y3 1G 1Y4 2G 2Y1 2G 2Y2 2G 2Y3 2G 2Y4	2 " " " " " " "	10 " " " " " " "	" " " " " " " "

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device types 02 and 03 1/ - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case R,S, X,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits		Unit
			Test no.	1G	1A1	2Y4	1A2	2Y3	1A3	2Y2	1A4	2Y1	GND	2A1	1Y4	2A2	1Y3	2A3	1Y2	2A4	1Y1	1/	V_{CC}		Min	Max	
10 T_C = 125°C	t_{PLH1} t_{PHL1} t_{PZL} t_{PZH} t_{PLZ} t_{PHZ}	Same tests and terminal conditions as subgroup 9 except T_C = 125°C.																							3 3 7 7 3 2	16/ 13 25 25 17/ 12	ns " " " " µs
11 T_C = -55°C	Same tests, terminal conditions and limits as subgroup 10 except T_C = -55°C.																										

1/ For type 02 - pin 19 is 2G.
for type 03 - pin 19 is 2$\overline{G}$.

2/ Apply 2.0 V for type 02; apply 0.8 V for type 03.

3/ Apply 0.5 V for type 02; apply 2.0 V for type 03.

4/ I_{IL} limits shall be as follows:

Symbol	Min/Max limits (µA) for circuit:		
	A	B	C
I_{IL}	0/-100	0/-100	0/-100

5/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather than short circuit current. The output conditions have been chosen to produce a current that closely approximates one half of the true short circuit output current I_{OS}.

6/ I_O limit for circuit C shall be -30 to -110 mA.

7/ Apply 5.5 V for type 02; apply GND for type 03.

8/ Apply GND for type 02; apply 5.5 V for type 03.

9/

	H	L	Z
02	17	28	32
03	15	24	27

10/ A ≥2.0 V and B ≤0.8 V.

11/ Apply A for type 02; apply B for type 03.

12/ Output voltages shall be either:

a. H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or

b. H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

13/ 11 ns for device type 02 and 10 ns for device type 03.

14/ 21 ns for device type 02 and 20 ns for device type 03.

15/ 15 ns for device type 02 and 13 ns for device type 03.

16/ 14 ns for device type 02 and 13 ns for device type 03.

17/ Max limit shall be 20 ns for type 02 and 18 ns for type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

1. Test condition D using the circuit shown on figure 3, or equivalent.

2. T_A = +125°C, minimum.

3. Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistics support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure and analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

f. Requirements for product assurance options.

g. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirement for JAN marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
V_{IN}	Voltage level at an input terminal
I_{IN}	Current flowing into an input terminal
t_{PHZ}	Output disable time from high level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high-impedance (off) state.
t_{PLZ}	Output disable time from low level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH}	Output enable time to high level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.
t_{PZL}	Output enable time to low level - The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.
I_{OZL}	Output current in the high-impedance mode with the output voltage low.
I_{OZH}	Output current in the high-impedance mode with the output voltage high.

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed Generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of Generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS240
02	54ALS241
03	54ALS244

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.

6.6 Manufacturer's designation. Manufacturer's circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturer's designation.

Device type	Circuits		
	A	B	C
	Texas Instrument	Motorola	National
01	X		
02	X		
03	X		

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1102)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:36Z
Check the source to verify that this is the current version before use.